

When Does Graham's Law Apply? Does Pressure Matter?

Question

- ▶ Does changing pressure prevent Graham's law from predicting effusion rates?

Evidence

- ▶ Effusion rate equals *number density* $\times$ molecular speed at the hole.
- ▶ Equal temperature gives the same average kinetic energy, so $v \propto 1/\sqrt{M}$.
- ▶ Different pressures produce different number densities.

Reasoning

- ▶ $\therefore$ rate depends on both density and speed, mass alone cannot determine effusion.
- ▶ $\therefore$ Graham's law requires *equal pressure* so molecular density is the same.
- ▶ $\therefore$ only *molar mass* remains to control the rate ratio.

Takeaway

- ▶ Equal pressure is essential for Graham's law.